



WHAT IS COLOUR GAMUT?

Colour gamut is the complete range of colours that a display system is capable of reproducing. Wider the colour gamut, greater the colour accuracy of the display. It is represented through the chromaticity diagram, where the RGB colour boundary denotes the human limit of colour perception and a TV's colour gamut is represented as a triangle compared to colour spaces such as sRGB, NTSC, and Adobe RGB. A good display should possess a colour gamut wide enough to be able to match or exceed the sRGB colour space.

glasses as such, and causes no harmful effect to the human eye

It's not really 3D

Most of the current generation's so called 3D display techniques do not actually produce a 3D image. Rather, what is produced is a pair of 2D images which can provide a stationary viewpoint that takes into account the depth of the visual. True-3D should also take into account the movement of the viewpoint and should provide visuals for several, if not all, possible viewing directions.

Even the most immersive option available nowadays, the

head mounted display, is prone to motion sickness and causes stress to the eye upon continuous usage due to the optical aids present in the viewing apparatus. On the opposite side when it comes to complexity level, Anaglyph systems are prone to chromatic aberration and often aren't able to provide true colour. Moreover, this kind of a system heavily relies on the assumption that both eyes of a person are able to function identically, while in real life, it is rarely so.

In Polarized displays, the screen is mostly shared between the two images with different polarization, causing an overall downgrade in resolution compared to full screen content. They are also known to make the image dimmer. Active shutter displays on the other hand require quite expensive and complicated viewing devices that need to be connected to the display for synchronization. And even though autostereoscopic systems remove the need for a viewing device, they restrict the viewing angle to a very specific area beyond which the view is distorted and no longer 3D.

Off late, holographic displays and true-3D displays using integral imaging and compressed light fields are emerging into the market. But just like 3D TVs, this technology is yet to establish its need to the consumer. 